

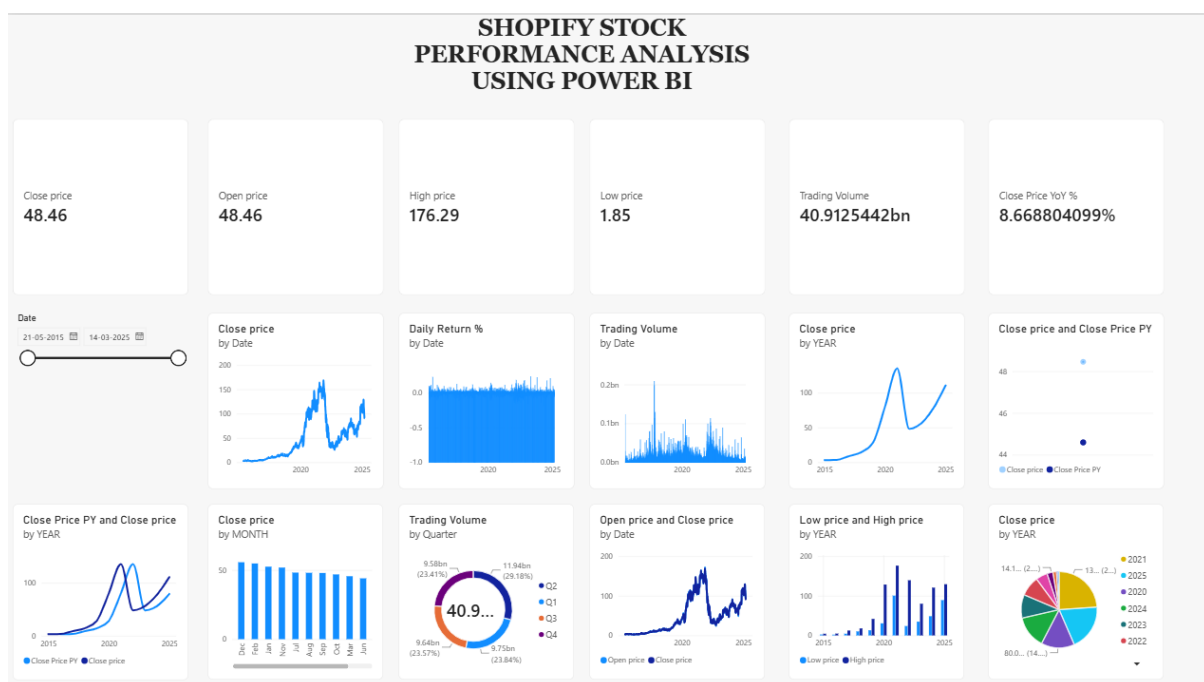
SHOPIFY STOCK PERFORMANCE ANALYSIS USING POWER BI

Tool Used: Microsoft Power BI

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Introduction:

This project focuses on analyzing Shopify stock market data using Microsoft Power BI. The dashboard provides an interactive view of stock prices, trading volume, daily returns and year-wise performance. Various charts, cards and slicers are used to identify trends and understand the movement of Shopify stock over time.



Objective of the Project:

The main objectives of this project are:

- To analyze Shopify stock price data.
- To study Open, High, Low and Close prices.
- To analyze trading volume.
- To calculate daily return percentage.

- To compare current closing price with previous-year closing price.
- To analyze monthly, quarterly and yearly trends.
- To create an interactive Power BI dashboard.

Dataset Description:

The dataset contains historical Shopify stock information. The major fields include Date, Open, High, Low, Close, Adjusted Close and Volume.

Queries [2]		= Table.TransformColumnTypes(#"Promoted Headers",{{"date", type datetimezone}, {"open", type number},					
SHOP_2015-05-21_2025...	Measure Table	date	1.2 open	1.2 high	1.2 low	1.2 close	
		Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	10
1		21-05-2015 12:00:00 AM -04:00	2.799999952	2.874000072	2.411000013	2.568	
2		22-05-2015 12:00:00 AM -04:00	2.607000113	3.109999895	2.599999905	2.83	
3		26-05-2015 12:00:00 AM -04:00	2.980000019	3.03399992	2.907999992	2.964	
4		27-05-2015 12:00:00 AM -04:00	3.066999912	3.08100009	2.700000048		
5		28-05-2015 12:00:00 AM -04:00	2.755000114	2.773999929	2.648000002	2.744	
6		29-05-2015 12:00:00 AM -04:00	2.724999905	2.737999916	2.687000036	2.720	
7		01-06-2015 12:00:00 AM -04:00	2.720000029	2.743999958	2.660000086	2.73	
8		02-06-2015 12:00:00 AM -04:00	2.746999979	2.746999979	2.664999962	2.694	
9		03-06-2015 12:00:00 AM -04:00	2.676000118	2.75	2.650000095	2.743	
10		04-06-2015 12:00:00 AM -04:00	2.742000103	2.749000072	2.551000118	2.677	
11		05-06-2015 12:00:00 AM -04:00	2.665999889	2.739000082	2.609999895	2.739	
12		08-06-2015 12:00:00 AM -04:00	2.690000057	2.747999907	2.681999922	2.697	
13		09-06-2015 12:00:00 AM -04:00	2.664000034	2.845999956	2.664000034	2.834	
14		10-06-2015 12:00:00 AM -04:00	2.845000029	2.888000011	2.795000076	2.809	
15		11-06-2015 12:00:00 AM -04:00	2.848999977	3.072999954	2.848999977	3.072	
16		12-06-2015 12:00:00 AM -04:00	3.131999969	3.369999886	3.125	3.362	
17		15-06-2015 12:00:00 AM -04:00	3.397000074	3.812000036	3.292000055	3.549	
18		16-06-2015 12:00:00 AM -04:00	3.605999947	3.837899923	3.584000111	3.811	
19		17-06-2015 12:00:00 AM -04:00	3.963000059	4.212999821	3.801000118	3.862	
20		18-06-2015 12:00:00 AM -04:00	3.842000008	3.888999939	3.432600021	3.555	
21		19-06-2015 12:00:00 AM -04:00	3.632999897	3.674000025	3.255000114	3.552	

Query Settings

PROPERTIES

Name

SHOP_2015-05-21_2025-03-16

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Column	Description
Date	Trading date
Open	Opening stock price
High	Highest price
Low	Lowest price
Close	Closing price
Adj Close	Adjusted closing price
Volume	Number of shares traded

Data Import and Cleaning:

The Shopify stock dataset was imported into Power BI. Power Query was used to inspect and prepare the data. Appropriate data types were assigned to date and numerical columns. The data was checked and prepared for further analysis.

Date Table Creation:

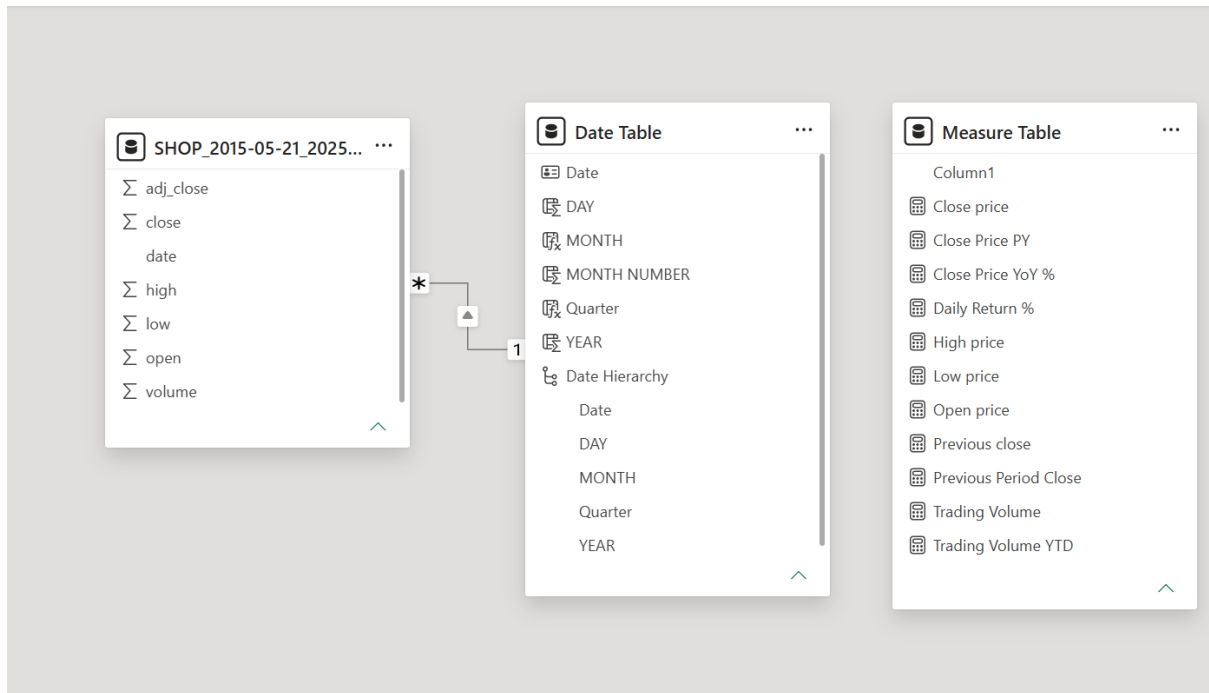
A separate Date Table was created to perform time-based analysis. The table contains Date, Day, Month, Month Number, Quarter and Year columns. This enables

monthly, quarterly and yearly analysis of stock performance.

1 Quarter = "Q" & FORMAT('Date Table'[Date], "Q")						
Date	Quarter	YEAR	MONTH	MONTH NUMBER	DAY	
21-05-2015 12:00:00 AM	Q2	2015	May	5	21	
22-05-2015 12:00:00 AM	Q2	2015	May	5	22	
23-05-2015 12:00:00 AM	Q2	2015	May	5	23	
24-05-2015 12:00:00 AM	Q2	2015	May	5	24	
25-05-2015 12:00:00 AM	Q2	2015	May	5	25	
26-05-2015 12:00:00 AM	Q2	2015	May	5	26	
27-05-2015 12:00:00 AM	Q2	2015	May	5	27	
28-05-2015 12:00:00 AM	Q2	2015	May	5	28	
29-05-2015 12:00:00 AM	Q2	2015	May	5	29	
30-05-2015 12:00:00 AM	Q2	2015	May	5	30	
31-05-2015 12:00:00 AM	Q2	2015	May	5	31	
01-06-2015 12:00:00 AM	Q2	2015	Jun	6	1	
02-06-2015 12:00:00 AM	Q2	2015	Jun	6	2	
03-06-2015 12:00:00 AM	Q2	2015	Jun	6	3	
04-06-2015 12:00:00 AM	Q2	2015	Jun	6	4	
05-06-2015 12:00:00 AM	Q2	2015	Jun	6	5	
06-06-2015 12:00:00 AM	Q2	2015	Jun	6	6	
07-06-2015 12:00:00 AM	Q2	2015	Jun	6	7	
08-06-2015 12:00:00 AM	Q2	2015	Jun	6	8	
09-06-2015 12:00:00 AM	Q2	2015	Jun	6	9	
10-06-2015 12:00:00 AM	Q2	2015	Jun	6	10	
11-06-2015 12:00:00 AM	Q2	2015	Jun	6	11	
12-06-2015 12:00:00 AM	Q2	2015	Jun	6	12	
13-06-2015 12:00:00 AM	Q2	2015	Jun	6	13	

Relationship Between Tables:

A relationship was created between the Date Table and Shopify Stock table using the Date column. The relationship enables the dashboard visuals and measures to perform accurate time-based analysis.



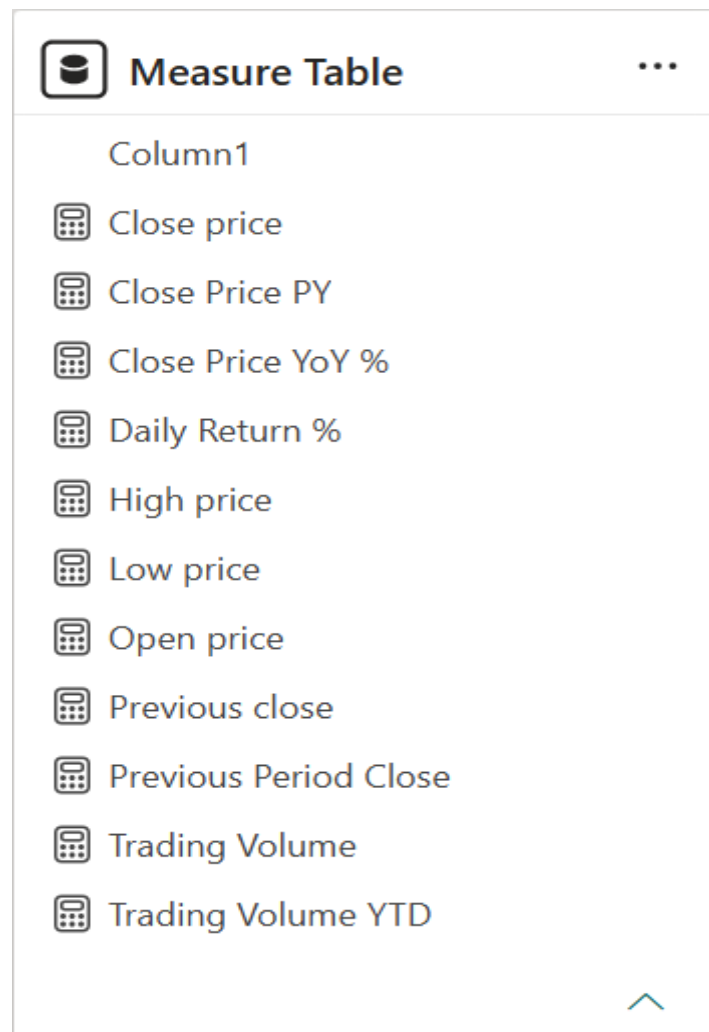
Measures Created:

Measures used:

- Close Price
- Open Price
- High Price
- Low Price
- Trading Volume
- Previous Close
- Daily Return %
- Previous Period Close
- Close Price PY

- Close Price YoY %

Several DAX measures were created to calculate important stock performance indicators. These measures were used in cards and different visualizations to make the dashboard interactive and informative.



Dashboard Visualizations:

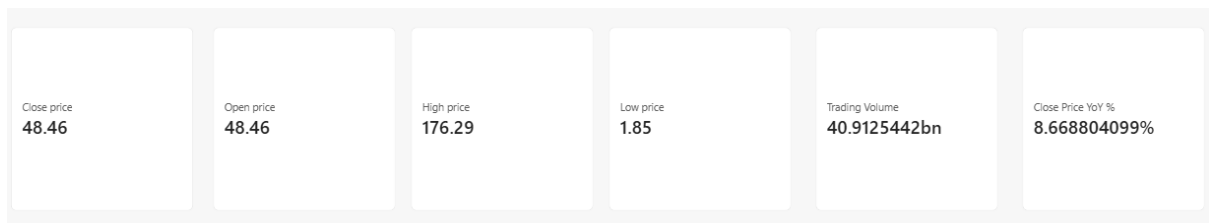
KPI Cards:

The dashboard contains KPI cards for:

- Close Price

- Open Price
- High Price
- Low Price
- Trading Volume
- Close Price YoY %

These cards provide a quick summary of important stock indicators.



Date Slicer:

A Date Slicer was added to allow users to select a specific date range. When the date range is changed, the dashboard visuals can be filtered accordingly.



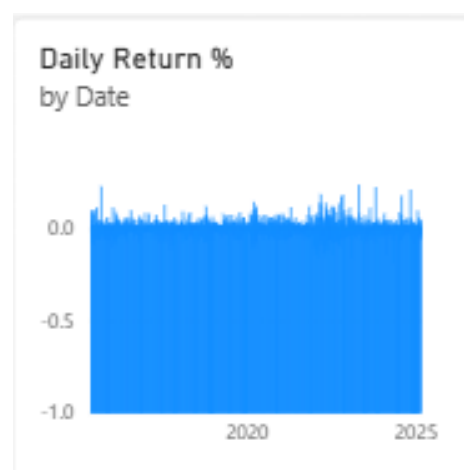
Close Price by Date:

The line chart shows the movement of Shopify's closing price over time. It helps to understand the overall trend and fluctuations in the stock price.



Daily Return %:

The Daily Return % chart represents the daily percentage change in the stock price. It helps to understand the day-to-day variation in stock performance.



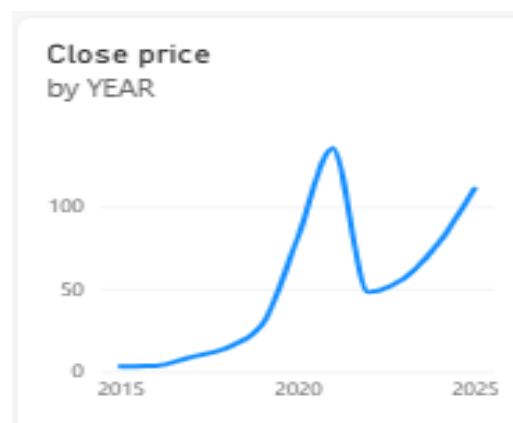
Trading Volume by Date:

This chart represents the trading volume over time. It helps identify periods with higher and lower trading activity.



Close Price by Year:

This line chart shows the closing price across different years. It helps in understanding long-term stock price movement.



Close Price vs Close Price PY:

This visualization compares the current closing price with the previous year's closing price. It provides a year-over-year comparison of stock performance.



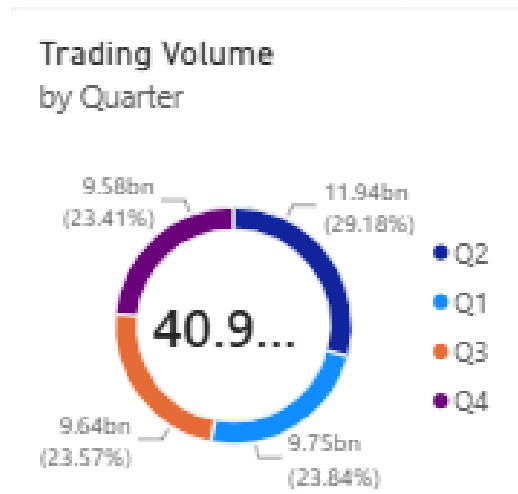
Close Price by Month:

This column chart displays the closing price for different months. It helps identify monthly variations in stock prices.



Trading Volume by Quarter:

The donut chart represents trading volume across different quarters. It provides a simple comparison of quarterly trading activity.



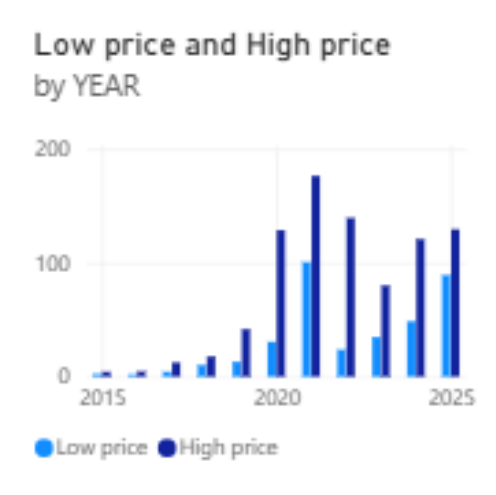
Open Price vs Close Price:

This line chart compares the opening and closing prices over time. It helps understand how the stock price changes during the trading period.



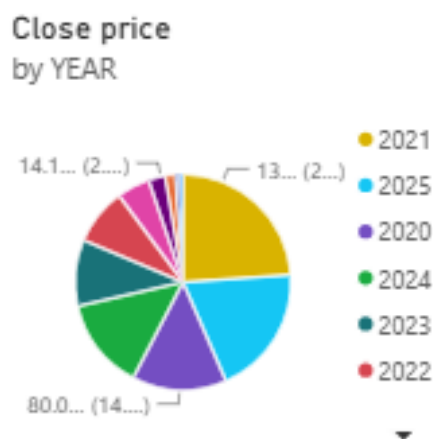
Low Price vs High Price:

This chart compares the low and high stock prices across different years. It helps understand the yearly price range.



Close Price by Year – Pie Chart:

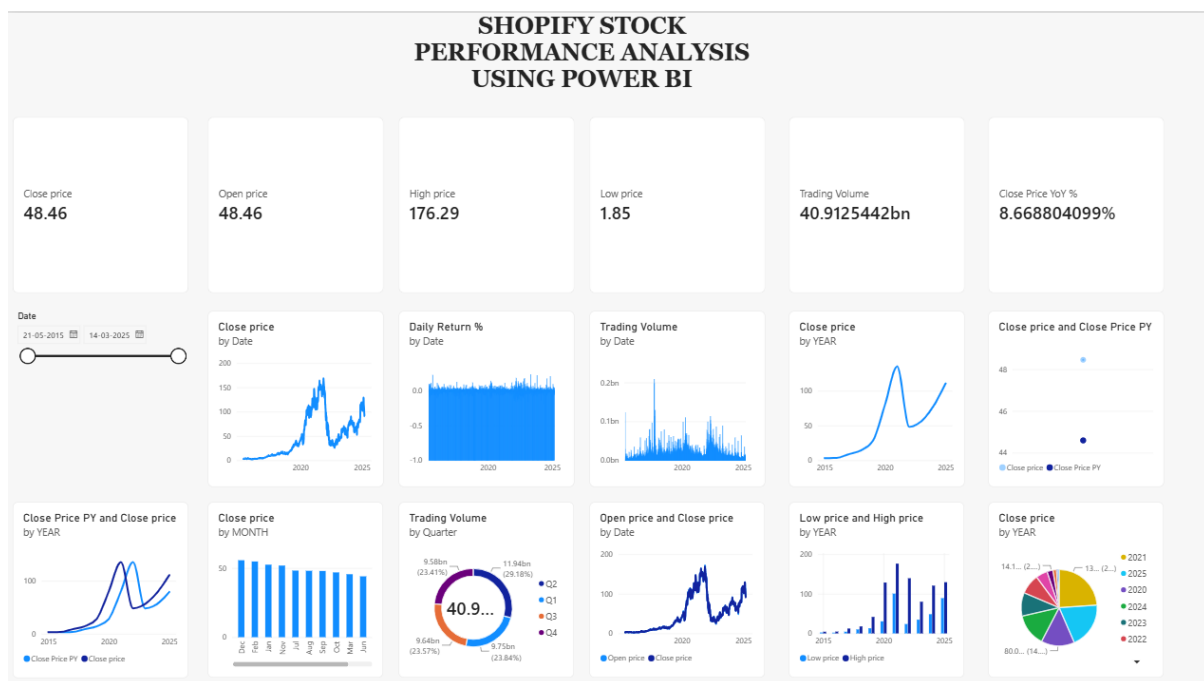
The pie chart provides a visual comparison of closing price values across different years.



Final Dashboard:

The final Power BI dashboard combines KPI cards, slicers, line charts, column charts and donut/pie charts into a single interactive report.

It provides a complete overview of Shopify stock performance and allows users to analyze the data from different time perspectives.



Key Observations:

From the dashboard analysis:

- Shopify's closing price changes significantly across different periods.
- Open and close prices can be compared using the line chart.

- High and low prices show the range of stock price movement.
- Trading volume varies across different dates and quarters.
- Daily Return % shows day-to-day changes in stock performance.
- Previous-year comparison provides an additional view of yearly performance.
- Monthly and quarterly visuals help understand time-based patterns.
- The Date Slicer makes the dashboard interactive.

Conclusion:

The Shopify Stock Performance Analysis project demonstrates how Microsoft Power BI can be used to transform raw stock data into an interactive dashboard. The project uses data cleaning, Date Table, relationships, DAX measures, KPI cards, slicers and different charts to analyze stock performance.

The dashboard provides a simple and effective way to understand Shopify stock prices, trading volume, daily returns and time-based performance.